

23009200111 陆文萱

$$1. (1) z = \frac{1}{3+2i} = \frac{3-2i}{13}$$

$$|z| = \sqrt{\left(\frac{3}{13}\right)^2 + \left(-\frac{2}{13}\right)^2} = \frac{\sqrt{13}}{13} \quad (|z_1 z_2| = |z_1| |z_2|)$$

$$x > 0, y < 0, \arg z = \arctan\left(-\frac{2}{3}\right)$$

$$\bar{z} = \frac{3+2i}{13}$$

$$(2) \frac{1}{i} - \frac{3i}{1-i} = -i - \frac{3i(1+i)}{2} = -i - \frac{3}{2}i + \frac{3}{2} = \frac{3}{2} - \frac{5}{2}i$$

$$|z| = \sqrt{\left(\frac{3}{2}\right)^2 + \left(-\frac{5}{2}\right)^2} = \frac{\sqrt{34}}{2}$$

$$x > 0, y < 0, \arg z = \arctan\left(-\frac{5}{3}\right)$$

$$\bar{z} = \frac{3}{2} + \frac{5}{2}i$$

$$2. (1) z = \frac{2i(-1-i)}{2} = 1-i = \sqrt{2}(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})) = \sqrt{2}e^{-\frac{\pi}{4}i}$$

$$(2) z = 2\sin^2\frac{\varphi}{2} + i \cdot 2\sin\frac{\varphi}{2}\cos\frac{\varphi}{2}$$

$$= 2\sin\frac{\varphi}{2}(\sin\frac{\varphi}{2} + i\cos\frac{\varphi}{2})$$

$$= 2\sin\frac{\varphi}{2}(\cos(\frac{\pi-\varphi}{2}) + i\sin\frac{\pi-\varphi}{2})$$

$$= 2\sin\frac{\varphi}{2}e^{i\frac{\pi-\varphi}{2}}$$

$$3. (1) -1 = \cos\pi + i\sin\pi$$

$$\omega_k = \sqrt[n]{r} e^{i\frac{\theta+2k\pi}{n}}$$

$$\therefore (\sqrt[6]{-1})_k = \cos\frac{\pi+2k\pi}{6} + i\sin\frac{\pi+2k\pi}{6} \quad (k=0, 1, 2, \dots, 5)$$

$$\text{当 } k=0 \text{ 时, } (\sqrt[6]{-1})_0 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$\text{当 } k=1 \text{ 时, } (\sqrt[6]{-1})_1 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$$

$$\text{当 } k=2 \text{ 时, } (\sqrt[6]{-1})_2 = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$\text{当 } k=3 \text{ 时, } (\sqrt[6]{-1})_3 = \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6} = -\frac{\sqrt{3}}{2} - \frac{1}{2}i$$

$$\text{当 } k=4 \text{ 时, } (\sqrt[6]{-1})_4 = \cos\frac{3}{2}\pi + i\sin\frac{3}{2}\pi = -i$$

$$\text{当 } k=5 \text{ 时, } (\sqrt[6]{-1})_5 = \cos\frac{11}{6}\pi + i\sin\frac{11}{6}\pi = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

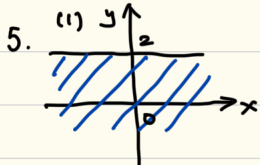
$$(2) 1+i = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$$

$$(1+i)^6 = (\sqrt{2})^6 \cos(\frac{\pi}{4} \cdot 6) + i\sin(\frac{\pi}{4} \cdot 6) = 8(\cos\frac{3}{2}\pi + i\sin\frac{3}{2}\pi) = -8i$$

$$4. 1 = \left(\frac{1+i}{1-i}\right)^n$$

$$\frac{1+i}{1-i} = \frac{(1+i)^2}{2} = i$$

$$\therefore 1 = (i)^n, n = 4k \quad (k \in \mathbb{N}_+)$$



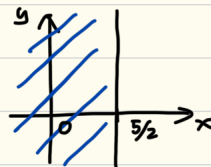
$$(2) \text{ 设 } z = x+yi$$

$$\text{显然 } z \neq 2.$$

$$|z-3| \geq |z-2|.$$

$$(x-3)^2 + (y)^2 \geq (x-2)^2 + (y)^2$$

$$\therefore x \leq \frac{5}{2}$$

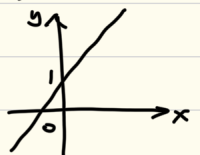


$$(3) \text{ 设 } z = x+yi.$$

$$z-i = x + (y-1)i$$

$$\tan\frac{\pi}{4} = \frac{y-1}{x}$$

$$\therefore y = x+1$$



$$6. \text{ 设 } z = \alpha+yi, \alpha = a+bi.$$

$$z\bar{z} + \alpha\bar{z} + \bar{\alpha}z + c = x^2 + y^2 + 2\operatorname{Re}(z\bar{\alpha}) + c$$

$$= x^2 + y^2 + 2(ax+by) + c$$

$$= (x-a)^2 + (y-b)^2 + (c-a^2-b^2) = 0 \quad (a^2+b^2 > c)$$

$$A(x^2+y^2) + Bx + Cy + D = 0$$

$$x = \frac{z+\bar{z}}{2}$$

$$y = \frac{z-\bar{z}}{2i}$$

7. (1) $z = 1 + yi$
 $w = \frac{1}{z} = \frac{1}{1+yi} = \frac{1-yi}{1+y^2}$
 $u = \frac{1}{1+y^2}, v = \frac{-y}{1+y^2}$
 $\therefore (u-\frac{1}{2})^2 + v^2 = \frac{1}{4}$

(2) $w = \frac{x-yi}{x^2+y^2}$ 又 $x^2+y^2 = 2x$
 $\therefore w = \frac{x-yi}{2x} = \frac{1}{2} - \frac{y}{2x}i$

? 这里 x 和 y 有范围吧
 $(-\frac{\sqrt{3}}{3} \leq \frac{y}{x} \leq \frac{\sqrt{3}}{3})$



8. 令 $z = x + iy, f(z) = u + iv$.

$\therefore f(z) = \frac{2xy}{x^2+y^2}$

令 $y = kx$.

$f(z) = \frac{2 \cdot x \cdot kx}{x^2 + k^2x^2} = \frac{2k}{1+k^2}$

$x \rightarrow 0$ 极限值随 k 值变化而变化.

$\therefore \lim_{z \rightarrow 0} f(z)$ 不存在.

法二: 令变点 $z = r(\cos\theta + i\sin\theta)$

$$f(z) = \frac{1}{2i} \cdot \frac{z^2 - \bar{z}^2}{2z\bar{z}} = \frac{1}{2i} \cdot \frac{(z+\bar{z})(z-\bar{z})}{r^2}$$

$$= \frac{1}{2ir^2} \cdot 2r\cos\theta \cdot 2ir\sin\theta$$

$$= 2\sin\theta \cos\theta = \sin 2\theta$$

沿正实轴 $\theta = 0, \lim_{z \rightarrow 0} f(z) = 0$. 沿 $\theta = \frac{\pi}{4}, \lim_{z \rightarrow 0} f(z) = 1$.

$\therefore f(z)$ 在原点无确定极限. 从而在 $z \rightarrow 0$ 原点不连续

9. (1) $f(z) = x^2 - iy$

$u = x^2, v = -y$

$u_x = 2x, u_y = 0, v_x = 0, v_y = -1$.

$u_x = v_y, x = -\frac{1}{2}; u_y = -v_x$

$\therefore$ 仅在直线 $x = -\frac{1}{2}$ 上, C.-R. 方程成立, 且偏导数连续,

$\therefore$ 仅在 $x = -\frac{1}{2}$ 上 $f(z)$ 可微, 在 z 平面上 $f(z)$ 处处不解析

$f'(z)|_{x=-\frac{1}{2}} = (u_x + iv_x)|_{x=-\frac{1}{2}} = -1$.

(2) $f(z) = \sinh x \cosh y + i \cosh x \sinh y$

$(\sinh x)' = \cosh x$

$u = \sinh x \cosh y, v = \cosh x \sinh y$

$(\cosh x)' = \sinh x$

$u_x = \cosh x \cosh y, u_y = \sinh x \sinh y, v_x = \sinh x \sinh y, v_y = \cosh x \cosh y$

$u_x = v_y, u_y = v_x$.

$\therefore f(z)$ 在 z 平面上处处可导, 处处解析.

10. $u = my^3 + nx^2y$ $v = x^3 + lxy^2$ $f'(z) = u_x + i v_x = u_x - i u_y = v_y + i v_x = v_y - i u_y$
 $u_x = ny \cdot 2x$, $u_y = m \cdot 3y^2 + nx^2$, $v_x = 3x^2 + ly^2$, $v_y = lx \cdot 2y$

$$\begin{cases} ny \cdot 2x = lx \cdot 2y \\ m \cdot 3y^2 + nx^2 = -(3x^2 + ly^2) \end{cases} \Rightarrow \begin{cases} m = 1 \\ n = -3 \\ l = -3 \end{cases}$$

 $f'(z) = u_x + i v_x = -6xy + i \cdot (3x^2 - 3y^2)$

11. (1) $\sin z = 0$ $\text{Ln} z = \ln|z| + i(\arg z + 2k\pi)$
 $\sin z = \frac{e^{iz} - e^{-iz}}{2i} = 0$

$\therefore e^{2iz} = 1$

$\therefore 2iz = \text{Ln} 1 = 0 + i(0 + 2k\pi)$, $k = 0, \pm 1, \pm 2, \dots$

$\therefore z = k\pi$ ($k \in \mathbb{Z}$)

(2) $1 + e^z = 0$

$z = \text{Ln}(-1) = \ln 1 + i(\pi + 2k\pi)$, $k = 0, \pm 1, \pm 2, \dots$

$\therefore z = (2k+1)\pi i$, $k = 0, \pm 1, \pm 2, \dots$

12. (1) 主值: $\ln z = \ln|z| + i \arg z$

$\ln(-i) = 0 + i \cdot (-\frac{\pi}{2}) = -\frac{\pi}{2}i$

(2) $\ln(-3+4i) = (\ln 5 + i(\arctan(-\frac{4}{3}) + \pi))$
 $= \ln 5 + i[\pi - \arctan \frac{4}{3}]$

(3. (1) $\exp \frac{1+i\pi}{4} = e^{\frac{1}{4}} e^{i\frac{\pi}{4}} = e^{\frac{1}{4}} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}) = \frac{\sqrt{2}}{2} e^{\frac{1}{4}} (1+i)$

(2) $3i = e^{i \text{Ln} 3} = e^{i(\ln 3 + 2k\pi)} = e^{-2k\pi} (\cos \ln 3 + i \sin \ln 3)$, $k = 0, \pm 1, \pm 2, \dots$

(14. $f(z) = \sqrt{z(z-1)(z-2)}$,

$n=2$, $N=3$, 可能的支点为 $0, 1, 2, \infty$.

$\therefore 2 \nmid 1, 2 \nmid 3$ bl.a. a 是被除数, b 是除数.

$\therefore$ 支点为 $0, 1, 2, \infty$.

$\therefore$ 取支割线为从原点出发的正实轴. 则 $f(z)$ 在沿此线割破 z 平面得到的区域 D 可以分出两个单值解析分支. z 从 -1 变动到 i 时,

$\Delta_c \arg f(z) = \frac{1}{2} [\Delta_c \arg z + \Delta_c \arg (z-1) + \Delta_c \arg (z-2)]$
 $= \frac{1}{2} [-\frac{\pi}{2} - \frac{\pi}{4} - \arctan \frac{1}{2}]$

$\therefore f(i) = |i(i-1)(i-2)|^{\frac{1}{2}} e^{i \Delta_c \arg f(z)} e^{i \arg f(z)}$
 $= \sqrt[4]{10} e^{i \cdot \frac{1}{2} \cdot [-\frac{7}{4}\pi - \arctan \frac{1}{2}]}$
 $= \sqrt[4]{10} e^{i\pi} \cdot e^{i \cdot \frac{1}{2} [+\frac{\pi}{4} - \arctan \frac{1}{2}]}$
 $= -\sqrt[4]{10} e^{\frac{i}{2} \arctan \frac{1}{2}}$

$\arg f(-1) = \arg(-\sqrt{6}i)$
 $= -\frac{\pi}{2}$
 $\tan(\frac{\pi}{4} - \arctan \frac{1}{2})$
 $= \frac{1 - \frac{1}{2}}{1 + i \cdot \frac{1}{2}} = \frac{1}{3}$

